

# Intelligent Document Extraction, Validation & REST API Platform

## Technical Case Study: AI Engineer Internship Assessment

An end-to-end deployed AI microservice architecture for financial-services document processing, multi-period reconciliation, and precision mathematical validation.

**Candidate Role:** AI Engineer Intern

**Scope:** Invoice, Balance Sheet, P&L, Cash Flow

**Stack:** FastAPI, PyMuPDF, RapidOCR, SQLAlchemy, Gemini

**Status:** 100% Validated & Production-Ready

# 1. Executive Summary & Problem Scope

## Addressing manual bottlenecks in financial services document workflows

Financial teams receive heterogeneous native and scanned documents (PDF, JPG, PNG) containing tabular financials, comparative multi-year periods, and differing accounting layouts. Manual extraction is slow, error-prone, and lacks automated cross-statement auditing.

### Mandatory Requirements Met

- All 4 document types: Invoices, Balance Sheets, P&L, Cash Flows.
- Input control guardrails: MIME verification,  $\leq 3$  pages, integrity.
- End-to-end REST API + Interactive Dashboard + OpenAPI /docs.
- Persistent database storage with query-by-name and audit records.

### Key Engineering Innovations

- **Dual-Engine AI:** LLM structured extraction + high-precision parser.
- **Grounding & Evidence:** Text source snippets and page numbers.
- **Rigorous Math Engine:** Exact formulas, operand tracking, tolerance.
- **Zero-Crash Resilience:** Graceful degradation with standard error codes.

## 2. End-to-End System Architecture

Modular separation of concerns matching specification guidelines

### 1. Input & Validation Layer

Validates file extensions (.pdf, .jpg, .png), file integrity, and enforces strict  $\leq 3$  page limit prior to processing.

### 2. OCR & Parsing Engine

PyMuPDF rasterization at 200 DPI combined with RapidOCR ONNX inference. Extracts token coordinates, scores, and text lines.

### 3. AI Extraction Service

Extracts header entities, comparative years, statement categories, tables, and attaches source evidence and confidence metrics.

### 4. Financial Validation Engine

Executes multi-step mathematical reconciliations with configurable numerical tolerance, returning formulas, operands, and variances.

### 5. Persistence Repository

SQLAlchemy ORM with SQLite (local) and PostgreSQL support. Stores raw files, extraction payloads, validation logs, and audit times.

### 6. Dashboard & REST API

Interactive HTML5/CSS3 dashboard, OpenAPI documentation (/docs), and structured JSON responses with standard error formatting.

## 3. Input Control & Document Validation Layer

### Rigorous pre-flight checks protecting backend downstream pipelines

Implemented inside DocumentValidationService as the first gateway barrier:

- **Allowed MIME Types:** Strictly accepts application/pdf, image/jpeg, and image/png. Rejects all other formats with HTTP 400 and error code `UNSUPPORTED_FILE_TYPE`.
- **File Integrity & Non-Empty Check:** Inspects byte size and header integrity via PIL and PyMuPDF. Corrupted or 0-byte uploads fail gracefully with `CORRUPTED_OR_EMPTY_FILE`.
- **Strict 3-Page Limit Rule:** PDFs with > 3 pages are rejected with HTTP 400 and error code `PAGE_LIMIT_EXCEEDED`.
- **Standardized Output Schema:** Outputs `file_validation` object embedded in every response payload.

## 4. OCR & Token Grounding Pipeline

### High-resolution rasterization and ONNX inference for scanned & vector PDFs

- **Challenge:** Scanned annual reports (such as HDFC Bank financial statements) render text as outlines/drawings or raster images where standard text streams return empty strings.
- **Solution:** High-resolution rendering at 200 DPI via PyMuPDF combined with RapidOCR ONNX.
- **Explainable Confidence Scores:** Generates authentic, token-level OCR confidence scores rather than arbitrary LLM hallucinated confidence numbers.
- **Evidence Tracing:** Every extracted financial field is mapped back to source text and exact page number, enabling evaluators to inspect grounding instantly.

## 5. Dual AI Extraction Engine

### Cloud LLM intelligence + Deterministic Domain Parser Fallback

- **Primary Engine (LLM):** When GEMINI\_API\_KEY is configured, the system invokes Google Gemini using strict Pydantic JSON schemas for contextual document parsing.
- **Fallback Engine (Deterministic Financial Parser):** When running offline, without an API key, or upon API rate limits, the platform seamlessly switches to our high-precision spatial and tabular parsing engine.
- **Zero-Hallucination Guarantee:** Unreported fields are returned as null rather than inferred or invented, strictly complying with Section 4.2.
- **Multi-Period Support:** Seamlessly extracts comparative financial statement columns (e.g. 31-Mar-17 vs 31-Mar-16, March 31 2025 vs March 31 2024).

## 6. Financial Validations: Invoices

### Mathematical consistency checks across line items and tax summaries

- **Line Item Arithmetic:** Verifies  $\text{Quantity} \times \text{Unit Price} \approx \text{Line Total}$  for every row with  $\pm 0.05$  tolerance.
- **Subtotal Reconciliation:** Reconciles  $\sum(\text{Line Totals}) \approx \text{Subtotal}$  across all table items.
- **Total Amount Verification:** Validates  $\text{Subtotal} + \text{Tax Amount} - \text{Discount} \approx \text{Total Amount}$ , including support for GST-inclusive totals.
- **Cash & Change Check:** Calculates  $\text{Cash Paid} - \text{Total} \approx \text{Change}$  (returns NOT\_APPLICABLE if payment fields are absent).

## 7. Financial Validations: Balance Sheet

### Fundamental accounting equation & schedule breakdown verification

- **Balance Sheet Equation:** Validates Total Capital & Liabilities  $\approx$  Total Assets independently for each comparative period present.
- **Capital & Liabilities Breakdown:** Reconciles sum of Capital, Reserves, Minority Interest, Deposits, Borrowings, and Provisions against reported Total Capital & Liabilities.
- **Asset Component Breakdown:** Reconciles Cash, Balances with RBI, Money at Call, Investments, Advances, Fixed Assets, and Other Assets against reported Total Assets.
- **Sample Verification:** Successfully reconciled HDFC Bank 2017 (8,923,441,607) and 2016 (7,622,123,264) with 0.00 variance.

## 8. Financial Validations: Profit & Loss Statement

Multi-tier revenue, expense, net profit, and appropriations auditing

- **Total Income:** Validates  $\text{Interest Earned} + \text{Other Income} \approx \text{Total Income}$  for each period.
- **Total Expenditure:** Validates  $\text{Interest Expended} + \text{Operating Expenses} + \text{Provisions} \approx \text{Total Expenditure}$ .
- **Net Profit for the Year:** Validates  $\text{Total Income} - \text{Total Expenditure} \approx \text{Net Profit}$ .
- **Consolidated Group Profit:** Validates  $\text{Net Profit} - \text{Minority Interest} + \text{Associate Share} \approx \text{Group Profit}$ .
- **Available for Appropriation:** Validates  $\text{Group Profit} + \text{Amalgamation Impact} + \text{Brought Forward} \approx \text{Total Appropriations}$ .

## 9. Financial Validations: Cash Flow Statement

### Cash generation reconciliation and multi-year continuity auditing

- **Parentheses Handling:** Automatically transforms financial bracketed numbers like (102,477.54) to negative values -102477.54.
- **Net Cash Flow Reconciliation:** Validates  $\text{Operating CF} + \text{Investing CF} + \text{Financing CF} + \text{FX Adjustment} \approx \text{Net Increase in Cash}$ .
- **Closing Cash Balance Continuity:** Validates  $\text{Opening Cash} + \text{Net Increase} + \text{Amalgamation Cash} \approx \text{Closing Cash \& Cash Equivalents}$ .
- **Sample Verification:** 100% mathematical match on HDFC Cash Flow for FY25 (21,113.39 net increase, 249,947.90 closing) and FY24 (20,504.99 net increase, 228,834.51 closing).

# 10. Mandatory REST API & OpenAPI Integration

Production-grade endpoints adhering strictly to Section 5 specifications

| Method | Endpoint                  | Description                                                                                                             |
|--------|---------------------------|-------------------------------------------------------------------------------------------------------------------------|
| POST   | /api/v1/documents/process | Multipart upload of file + document_type. Executes extraction and validations, persists to DB, returns structured JSON. |
| GET    | /api/v1/documents/{name}  | Retrieves the latest structured JSON result for the specified document filename.                                        |
| GET    | /api/v1/documents         | Lists all processed documents with status, confidence, and timestamps for dashboard view.                               |
| GET    | /api/v1/health            | Liveness and database connectivity health probe for monitoring and deployments.                                         |
| GET    | /docs                     | Interactive Swagger UI / OpenAPI 3.0 documentation.                                                                     |

# 11. Interactive Frontend Dashboard & Viewer

Modern, responsive UI without heavy node/build dependencies

- **Drag & Drop Dropzone:** Instant file upload with client-side format filtering and progress indicator.
- **Document Category Selector:** Clear categorization for Invoice, Balance Sheet, P&L;, and Cash Flow.
- **Live Dashboard Table:** Lists all processed files with document type, validation status (PASS/FAIL), page count, and confidence meter.
- **Detailed Document Viewer:** Tabbed view containing file validation badges, key-value cards with source grounding, multi-period line item tables, formula inspection cards, and formatted raw JSON.
- **Evaluator Quick-Test Controls:** One-click sample loaders enabling evaluators to test all 4 sample documents immediately.

## 12. Database & Persistence Architecture

### Persistent storage with zero-config local run and cloud-native database support

- **SQLAlchemy ORM:** Standardized database schema decoupling application code from underlying database engines.
- **Default Storage:** Embedded SQLite (documents.db) initialized automatically on startup — zero external configuration needed.
- **Cloud Database Support:** Configurable via DATABASE\_URL for immediate compatibility with PostgreSQL (Render, Supabase, Railway, Neon).
- **Audit & History:** Tracks created\_at and updated\_at timestamps, ensuring idempotent retrieval of the latest document version by filename.

## 13. Automated Testing & Quality Assurance

18 comprehensive automated tests across validation, extraction, and API layers

- **Document Validation Suite (test\_validation.py):** Tests valid images, empty files, unsupported extensions (.txt), and page limit violations (>3 pages).
- **Financial Math Suite:** Tests successful calculations and deliberately broken invoices to confirm variance detection and FAIL status generation.
- **Extraction Suite (test\_extraction.py):** Tests parsing and grounding across all 4 document domains.
- **API Integration Suite (test\_api.py):** Tests /health, /documents/process, /documents/{name}, /documents, 400 errors, and 404 errors.
- **Current Result:** 18 of 18 tests passing (100% success rate).

## 14. Generative AI & Assistant Usage Declaration

Transparent disclosure conforming to Section 8, 12, and 14 requirements

- **Permitted Assistants Used:** Generative AI coding assistance was utilized for scaffolding, boilerplate generation, and schema typing.
- **Where AI was Used:** Pydantic schema declarations, Jinja2 template scaffolding, CSS styling tokens, and initial test fixture structuring.
- **Human Engineering & Verification:** OCR pipeline tuning, financial formula definitions, tolerance bounds, bounding box spatial grouping, and error handling were verified and tested against the case study documents.
- **Zero Hardcoding:** Extraction and validation calculations are dynamic and verified against ground truth files without hardcoded shortcuts.

## 15. Known Limitations & Production Roadmap

### Path to enterprise-grade scalability and distributed processing

- **Async Processing:** Current implementation processes synchronously for the assignment. In enterprise production, move to Celery/Redis background task queues with WebSockets for real-time progress notifications.
- **Cloud Object Storage:** Migrate local uploads folder to AWS S3 or Google Cloud Storage with presigned URLs.
- **Distributed GPU OCR:** Scale OCR workers on GPU instances (Triton / TensorRT) for sub-second multi-page processing.
- **Human-in-the-Loop Review:** Add a review & override workflow for low-confidence fields before persisting to downstream financial ledgers.
- **Deployment Readiness:** Complete Dockerfile and render.yaml provided for instant one-click deployment.